

Lab — Movie Review Sentiment Analyser

Part A — Build the review dataset

```
import pandas as pd
from transformers import pipeline
reviews_data = [
    # Bollywood
    {'movie': '3 Idiots', 'source': 'Bollywood',
     'review': 'An absolute masterpiece! Makes you laugh and cry. '
     'Aamir Khan delivers one of his finest performances ever.'},
    {'movie': '3 Idiots', 'source': 'Bollywood',
     'review': 'Overrated and preachy. The message was fine but '
     'execution felt too theatrical and unrealistic.'},
    {'movie': 'Dangal', 'source': 'Bollywood',
     'review': 'Incredibly inspiring! Geeta and Babita brought tears '
     'to my eyes. Best sports film ever made in India.'},
    {'movie': 'Dangal', 'source': 'Bollywood',
     'review': 'Good film but very long. Some scenes dragged unnecessarily.'},
    # Hollywood
    {'movie': 'Inception', 'source': 'Hollywood',
     'review': 'Mind-bending and visually stunning. Nolan at his absolute best.'},
    {'movie': 'Inception', 'source': 'Hollywood',
     'review': 'Too complicated. I had no idea what was happening half the time.'},
    {'movie': 'Parasite', 'source': 'Hollywood',
     'review': 'Perfect film. The class commentary is brilliant. Deserved every award.'},
    {'movie': 'Parasite', 'source': 'Hollywood',
     'review': 'I expected more from an Oscar winner. Ending felt rushed.'},
]
df = pd.DataFrame(reviews_data)
```

Part B — Analyse sentiment

```
# Load once – not inside a loop (that would reload weights every iteration)
sentiment = pipeline('sentiment-analysis',
model='distilbert-base-uncased-finetuned-sst-2-english')
# Batch all reviews at once – faster than one-by-one
results = sentiment(df['review'].tolist())
df['sentiment'] = [r['label'] for r in results] # POSITIVE / NEGATIVE
df['confidence'] = [r['score'] for r in results] # 0.0 to 1.0
# sentiment_score: 1.0 = very positive, 0.0 = very negative
df['sentiment_score'] = df['confidence'].where(
df['sentiment'] == 'POSITIVE',
1 - df['confidence']
)
print(df[['movie','sentiment','confidence']].to_string(index=False))
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: User
The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (
You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access publ
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated req

config.json: 100% 629/629 [00:00<00:00, 28.8kB/s]

model.safetensors: 100% 268M/268M [00:02<00:00, 125MB/s]

Loading weights: 100% 104/104 [00:00<00:00, 2690.76it/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.97kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 6.55MB/s]

movie	sentiment	confidence
3 Idiots	POSITIVE	0.999879
3 Idiots	NEGATIVE	0.999016
Dangal	POSITIVE	0.999878
Dangal	NEGATIVE	0.992343
Inception	POSITIVE	0.999849
Inception	NEGATIVE	0.999590
Parasite	POSITIVE	0.999857
Parasite	NEGATIVE	0.998913

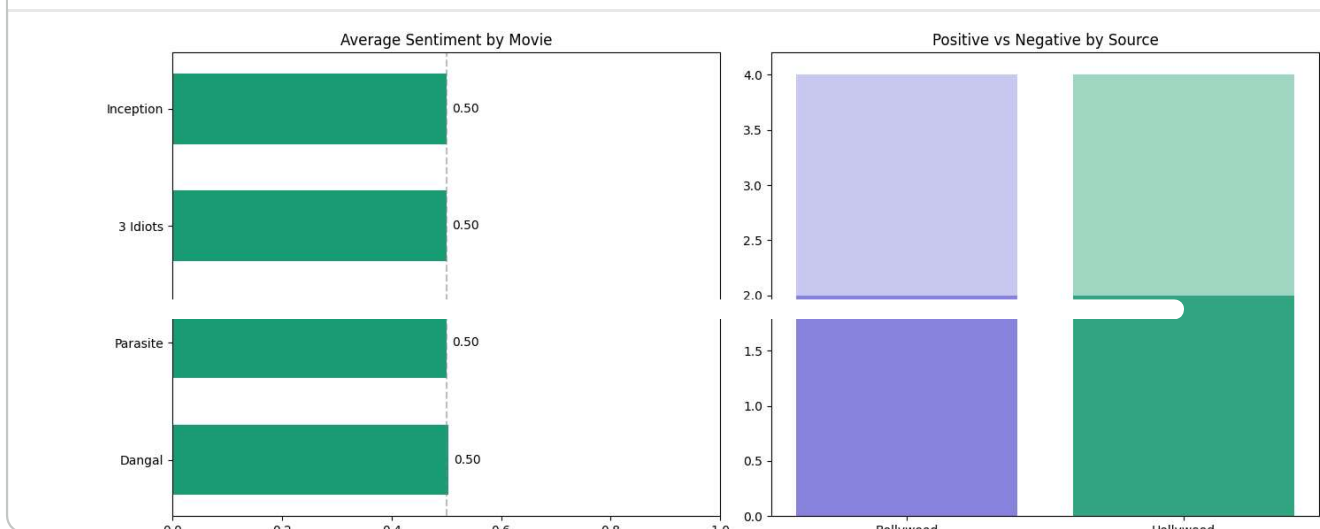
Part C — Visualise results

```
import matplotlib.pyplot as plt
fig, axes = plt.subplots(1, 2, figsize=(14, 6))
# Chart 1: Average sentiment score per movie
ax1 = axes[0]
movie_scores = df.groupby('movie')['sentiment_score'].mean().sort_values(ascending=True)
colours = ['#1D9E75' if v >= 0.5 else '#D85A30' for v in movie_scores.values]
bars = ax1.barh(movie_scores.index, movie_scores.values, color=colours,
```

```

height=0.6)
ax1.set_xlim(0, 1)
ax1.axvline(0.5, color='gray', linestyle='--', alpha=0.5)
ax1.set_title('Average Sentiment by Movie')
for bar, val in zip(bars, movie_scores.values):
    ax1.text(val+0.01, bar.get_y()+bar.get_height()/2, f'{val:.2f}',
             va='center')
# Chart 2: Positive vs Negative split by source
ax2 = axes[1]
for source, colour in [('Bollywood', '#7F77DD'), ('Hollywood', '#1D9E75')]:
    subset = df[df['source'] == source]
    pos = (subset['sentiment']=='POSITIVE').sum()
    neg = (subset['sentiment']=='NEGATIVE').sum()
    ax2.bar(source, pos, color=colour, alpha=0.9)
    ax2.bar(source, neg, bottom=pos, color=colour, alpha=0.4)
    ax2.set_title('Positive vs Negative by Source')
plt.tight_layout()
plt.savefig('sentiment_analysis.png', dpi=150, bbox_inches='tight')
plt.show()

```



Part D — Predict on new user input

```

def analyse_review(review_text, movie_name='Unknown'):
    result = sentiment(review_text)[0]
    label = result['label']
    conf = result['score']
    if label == 'POSITIVE':
        score = conf
        verdict = 'LOVED IT' if conf > 0.95 else 'LIKED IT'
    else:
        score = 1 - conf
        verdict = 'HATED IT' if conf > 0.95 else 'DISLIKED IT'
    print(f'Movie : {movie_name}')
    print(f'Verdict : {verdict}')
    print(f'Score : {score:.1%} positive')
    print(f'Review: {review_text[:80]}...')

```

```
return {'movie': movie_name, 'verdict': verdict, 'score': score}

analyse_review(
    'Rajkummar Rao was phenomenal. A raw, honest performance. '
    'Easily the best Hindi film of the decade.',
    'Stree'
)
```

Movie : Stree
Verdict : LOVED IT
Score : 100.0% positive
Review : Rajkummar Rao was phenomenal. A raw, honest performance. Easily the bes
{'movie': 'Stree', 'verdict': 'LOVED IT', 'score': 0.9998749494552612}

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